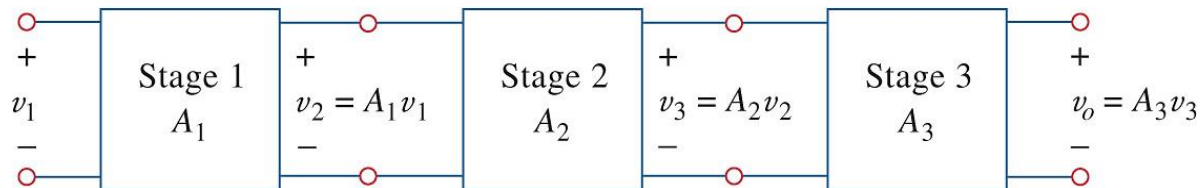


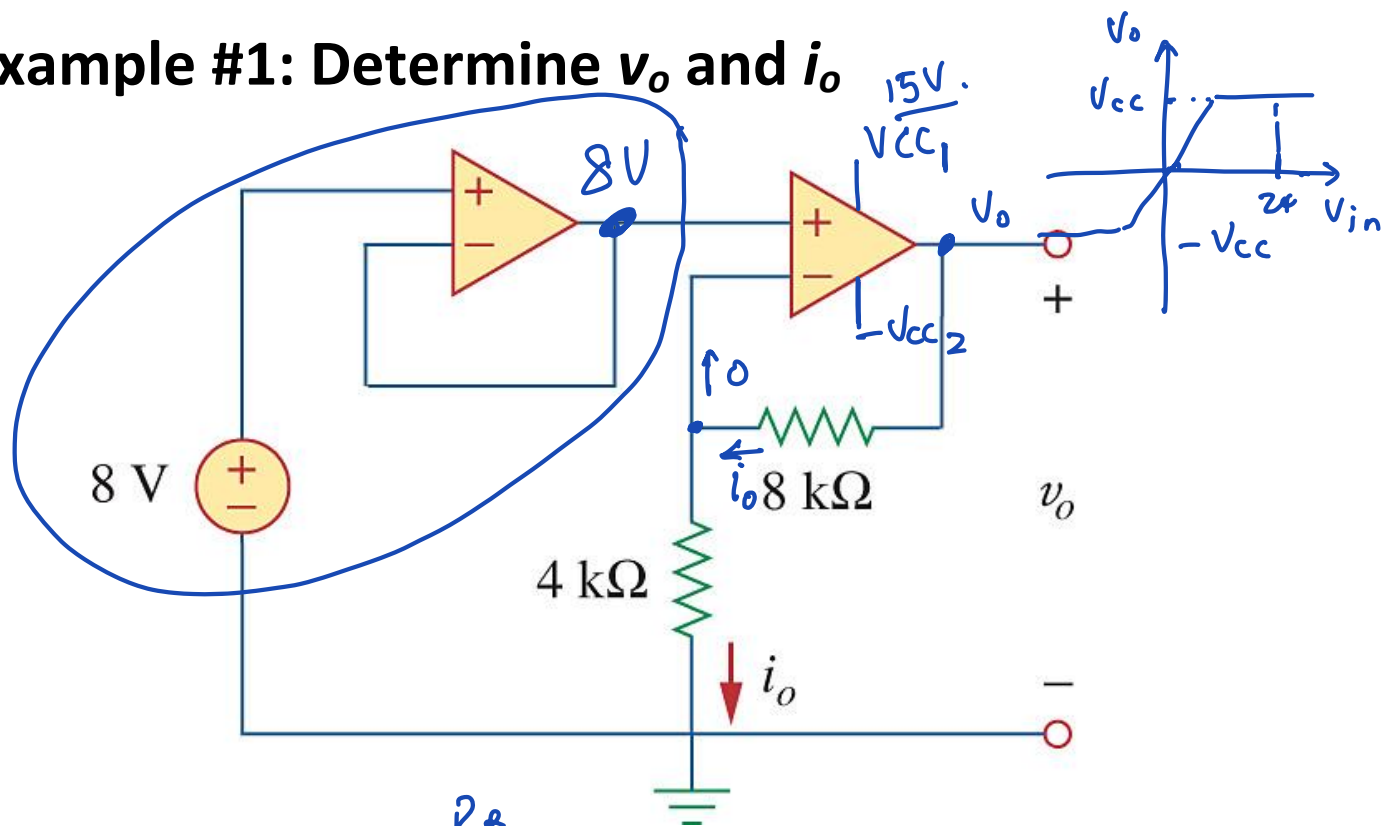
L#21: Cascaded Op Amp Circuit

Two or more Op Amps with “head-to-tail” connections



$$\underline{V_o} = A_3 V_3 = A_3 \underline{A_2 V_2} = A_3 A_2 A_1 \underline{V_1}$$

Example #1: Determine v_o and i_o

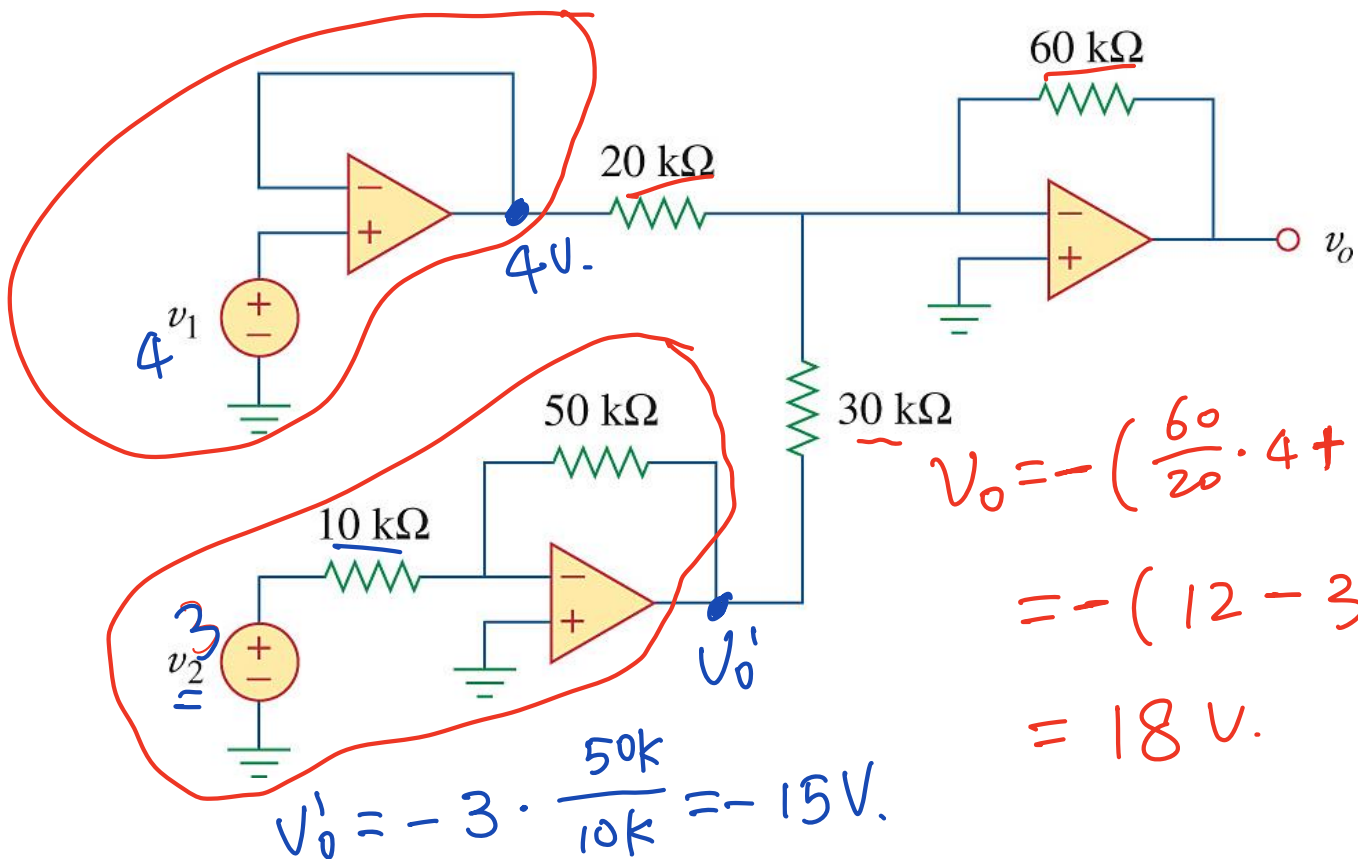


$$v_o = v_i \left(1 + \frac{R_f}{R_1} \right)$$

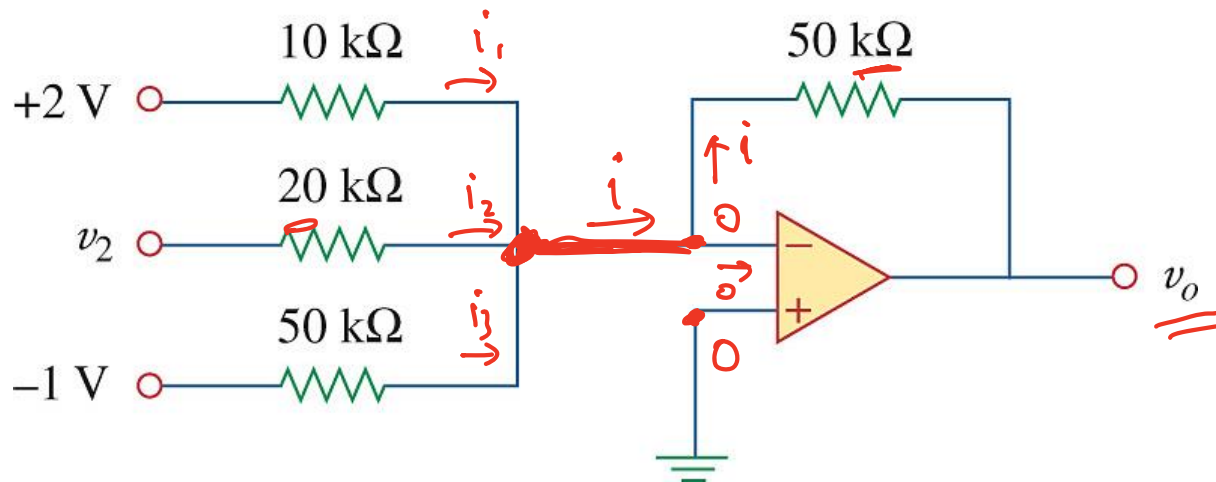
$$= 8 \cdot \left(1 + \frac{8k}{4k} \right) = \underline{24V}$$

$$i_o = \frac{v_o}{(8+4)k} = 2 \text{ mA}$$

Example #2: If $v_1 = 4\text{ V}$ and $v_2 = 3\text{ V}$, determine v_o .



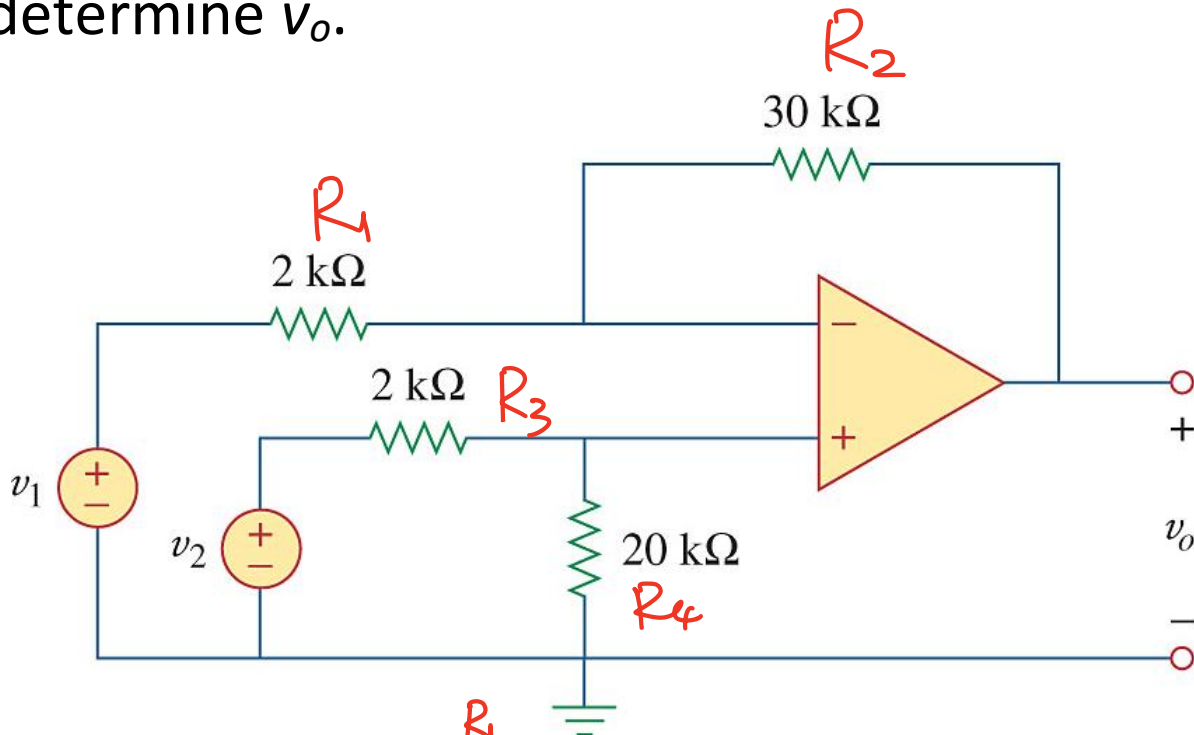
Example #3: Determine v_2 such that $v_o = -16.5$ V.



$$\begin{aligned}
 i &= i_1 + i_2 + i_3 \\
 &= \frac{2 - 0}{10\text{K}} + \frac{v_2 - 0}{20\text{K}} + \frac{-1 - 0}{50\text{K}} = \frac{0 - (-16.5)}{50\text{K}}
 \end{aligned}$$

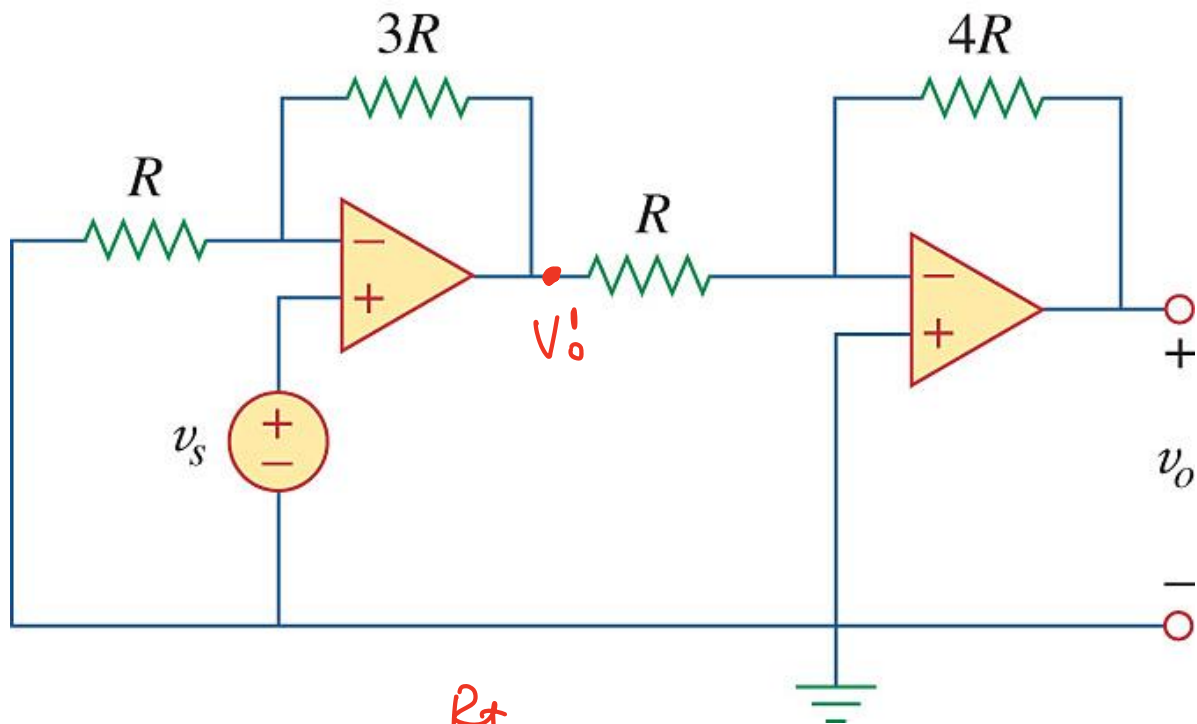
v_o

Example #4: If $v_1 = 1\text{ V}$ and $v_2 = 2\text{ V}$, determine v_o .



$$\begin{aligned}
 V_o &= \frac{R_2}{R_1} \cdot \frac{1 + \frac{R_1}{R_2}}{1 + \frac{R_3}{R_4}} \cdot V_2 - \frac{R_2}{R_1} \cdot V_1 \\
 &= \frac{30}{2} \cdot \frac{1 + \frac{2}{30}}{1 + \frac{2}{20}} \cdot 2 - \frac{30}{2} \cdot 1 \\
 &= 30 \cdot \frac{1 + \frac{2}{30}}{1.1} - 15 \\
 &= \frac{32}{1.1} - 15 > 0.
 \end{aligned}$$

Example #5: Determine v_o/v_s if $R = 20 \text{ k}\Omega$.

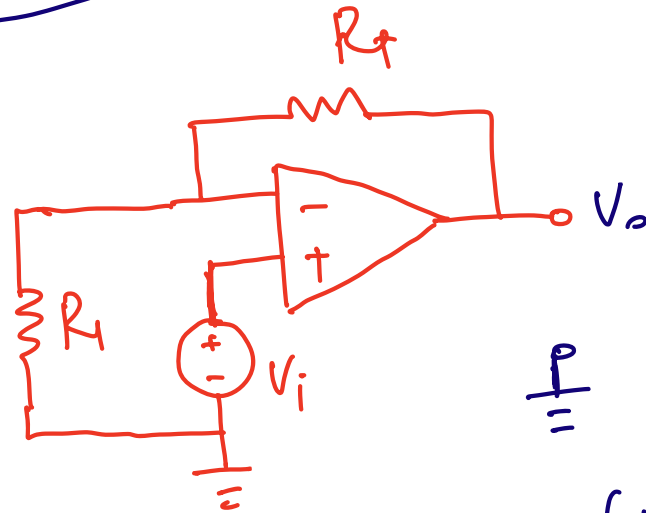
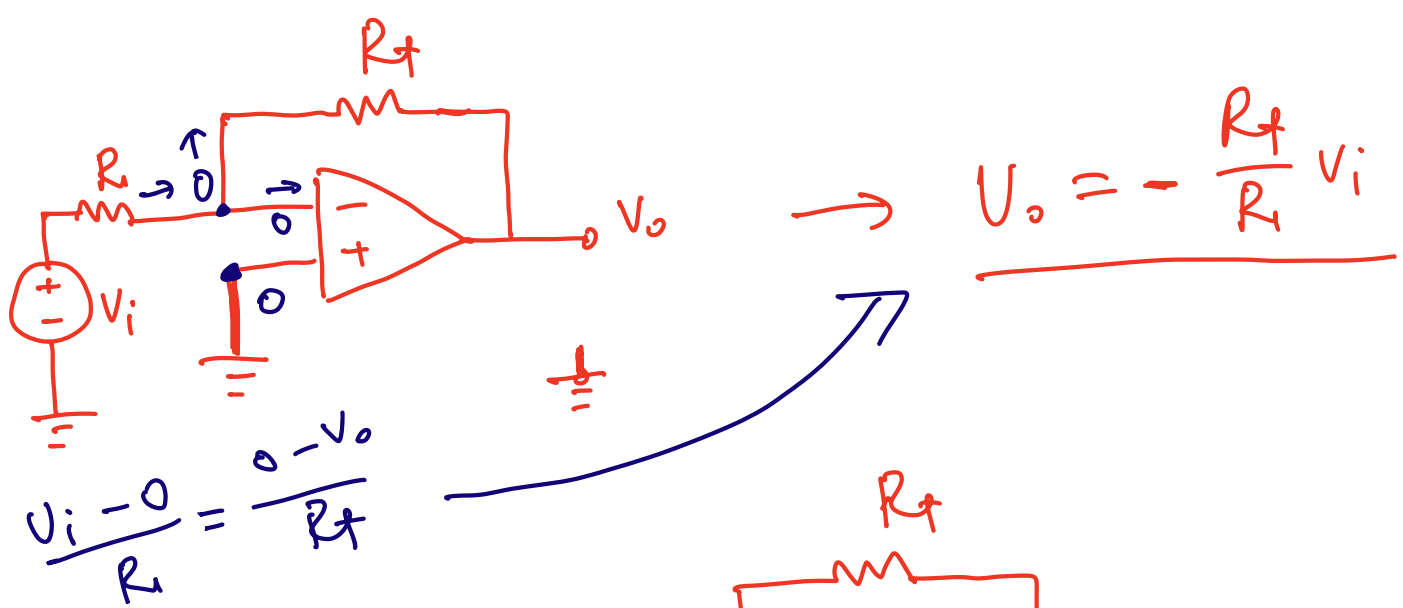


$$V_o' = V_s \left(1 + \frac{R_f}{R_i} \right)$$

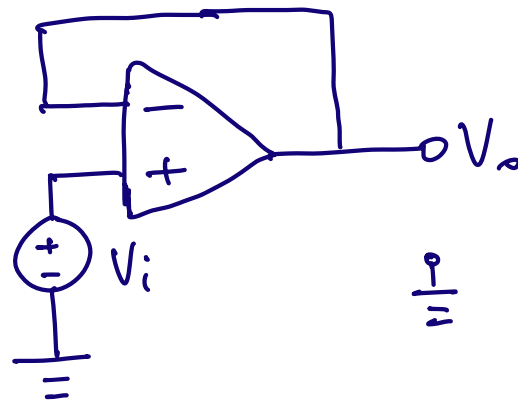
$$= V_s (1 + 3) = 4V_s$$

$$V_o = - \frac{R_f}{R_i} \cdot V_o' = - \frac{4R}{R} \cdot 4V_s = -16V_s.$$

$$\frac{V_o}{V_s} = -16.$$

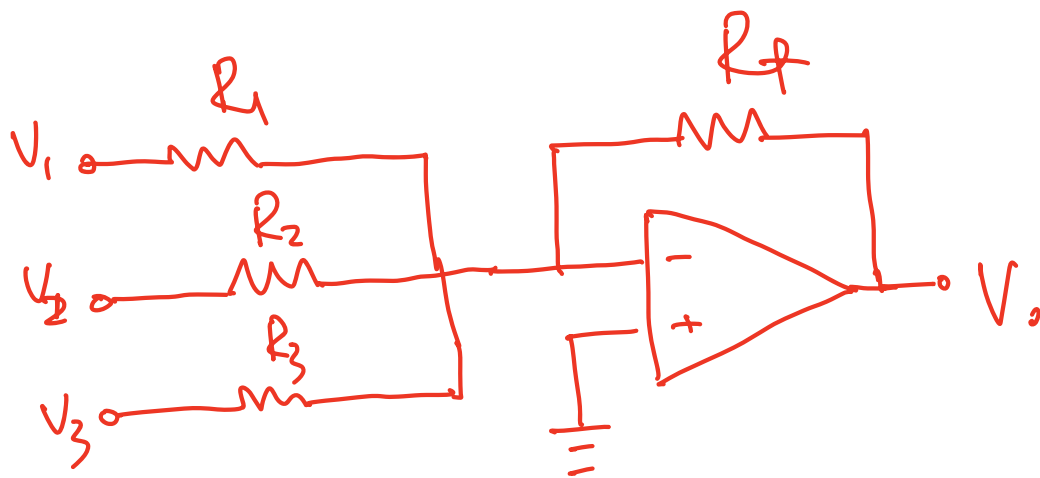


Let $R_f \rightarrow 0$
 $R_i \rightarrow \infty$

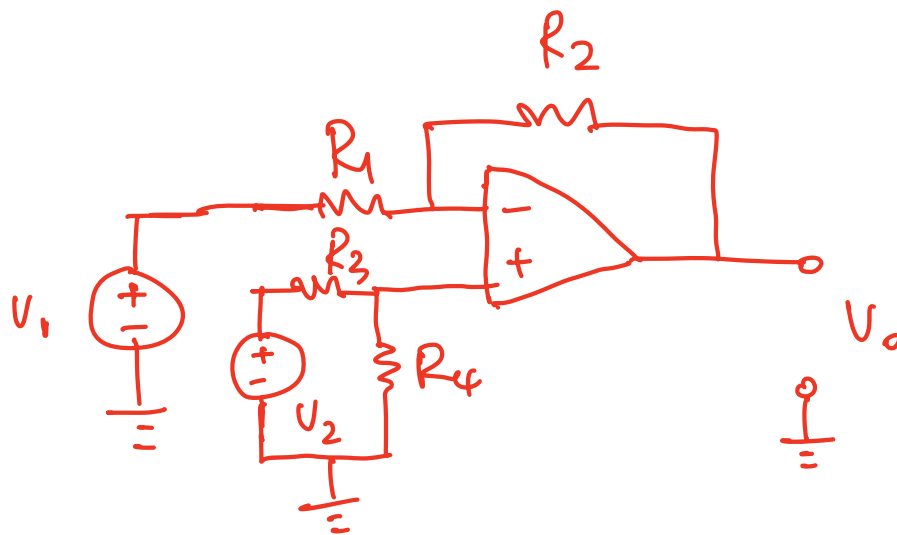


$$V_o = V_i \left(1 + \frac{0}{\infty}\right)$$

$$V_o = V_i$$



$$V_o = - \left(\frac{R_f}{R_1} V_1 + \dots + \frac{R_f}{R_3} V_3 \right).$$

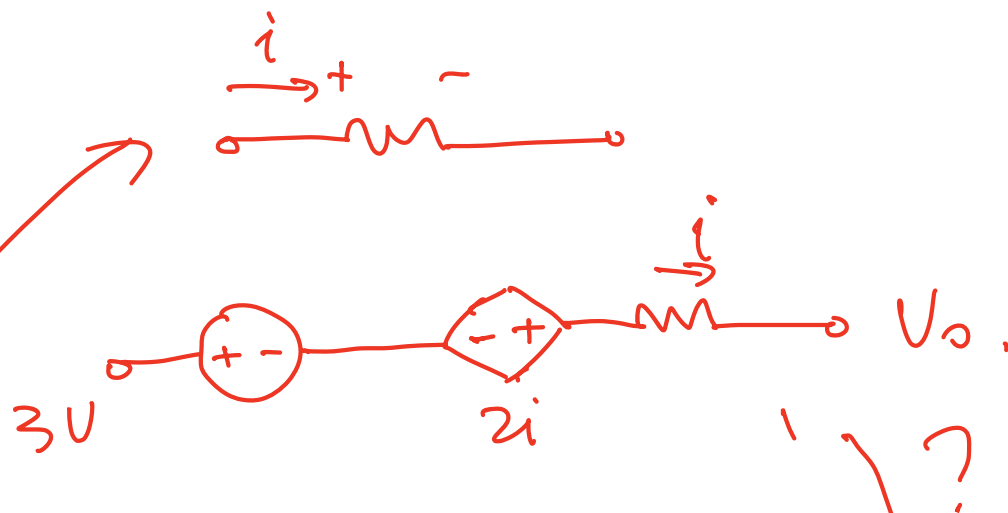


$$V_o = V_2 \underbrace{\frac{R_4}{R_3 + R_4}}_{\text{equivalent input}} \underbrace{\left(1 + \frac{R_2}{R_1} \right)}_{\text{non-inverting op Amp}} - \frac{R_2}{R_1} V_1.$$

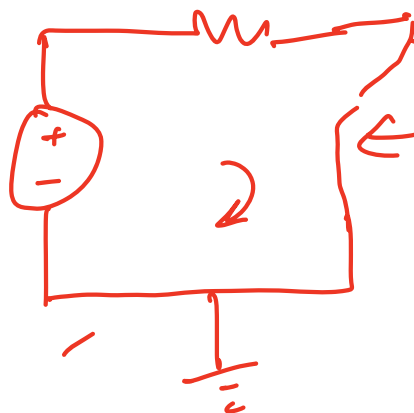
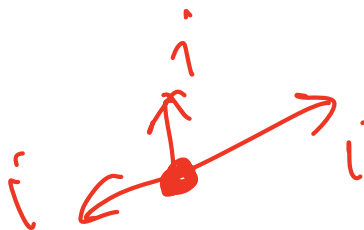
equivalent
input

non-inverting
op Amp.

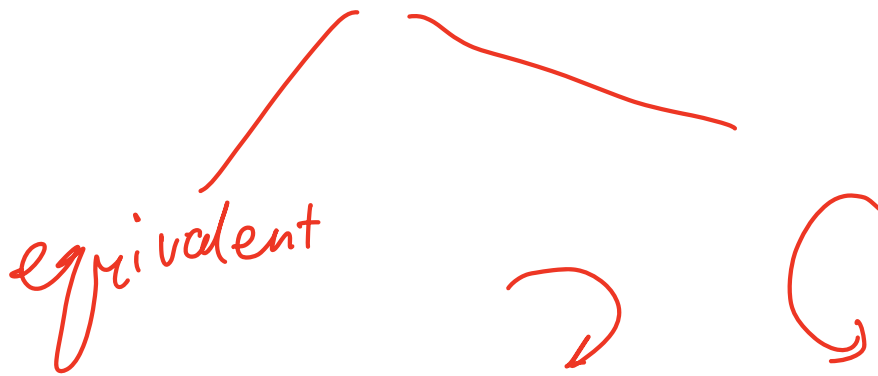
- ??
- unit, V , I , P , E
 - Ohm's law.



- KCL, KVL.



- mesh. node.
super mesh, super node.
inspection.
- transistors.



Ch 4: Superposition.
Source trans.
Norton, Thevenin.
max power.